BLUESTOCK MUTUAL FUND ANALYTICS DASHBOARD

Final Analytics Report
Bluestock Fintech Internship — Capstone Project

Detail	Information
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81IL Cr **Industry Total** **Total**	**31k cr** **Monthly Record** **SIP Inflow**	**26.12cr** **Active investor** **Total Folios**	1908 **Active Fundas** **schemes**

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1. Executive Summary

This report presents a comprehensive analytics study of the Indian Mutual Fund industry conducted as part of the Bluestock Fintech Internship Capstone Project. Over 7 days, a complete data analytics pipeline was designed and implemented — from raw data ingestion to interactive Power BI dashboards and advanced risk metrics.

Key Objective: Build a production-ready MF analytics platform covering 40 funds, 32,778 investor transactions, and ■81 Lakh Crore in total AUM.

1.1 Project Highlights

- Analyzed 40 mutual fund schemes across 10 major AMCs
- Processed 32,778 investor transactions from Jan 2022 to May 2026
- Built a 4-page interactive Power BI dashboard
- Computed risk metrics: VaR, CVaR, Sharpe Ratio, Beta, Alpha
- Identified 1,332 at-risk SIP investors (gap > 35 days)
- Built an AI fund recommender based on risk appetite

1.2 Key KPIs

81IL Cr **Industry Total** **Total**	**31k cr** **Monthly Record** **SIP Inflow**	**26.12cr** **Active investor** **Total Folios**	1908 **Active Fundas** **schemes**

1.3 Summary of Findings

Finding	Insight	Impact
Highest VaR Fund	Small Cap: -2.7%/day	High Risk
Best Sharpe Ratio	ICICI Pru Liquid: 7.68	Excellent
SIP Continuity	Only 2.2% regular investors	Critical
Top Cohort	2024: 4,624 investors	High Value
Most Concentrated	Axis Bluechip HHI: 2,064	Sector Risk
Top Category FY25	Liquid: ■4.5L Cr inflow	Strong

2. Data Sources & Overview

The project uses 10 structured CSV datasets covering all dimensions of the Indian Mutual Fund ecosystem — from fund metadata to investor transactions, NAV history, and benchmark indices.

#	Dataset	Description	Rows	Key Columns
1	01_fund_master.csv	Fund metadata: name, AMC, category, plan	40	amfi_code, scheme_name, category
2	02_nav_history.csv	Daily NAV for all 40 funds (2022-2026	46,000	amfi_code, date, nav
3	03_aum_by_fund_house.csv	Monthly AUM per AMC	90	fund_house, date, aum_crore
4	04_monthly_sip_inflows.csv	Monthly SIP inflow data	48	month, sip_inflow_crore
5	05_category_inflows.csv	Net inflows by fund category	144	category, month, net_inflow_crore
6	06_industry_folio_count.csv	Total folios per month	21	month, total_folios_crore
7	07_scheme_performance.csv	Returns, Sharpe, Alpha, Beta	40	amfi_code, return_3yr_pct, sharpe_ratio
8	08_investor_transactions.csv	Individual investor transactions	32,778	investor_id, amount_inr, state
9	09_portfolio_holdings.csv	Fund stock holdings & weights	322	amfi_code, sector, weight_pct

2.1 Data Quality Summary

Before analysis, all datasets were thoroughly cleaned and validated. The following quality checks were applied:

- Duplicate removal: Deduplicated on composite keys (amfi_code + date)
- Null handling: Dropped rows with null values in critical numeric columns
- Type conversion: Dates parsed to datetime, numerics cast with error coercion
- AUM scaling: Scaled to match real-world ■81L Cr industry total
- SIP calibration: Latest month set to ■31,000 Cr (record inflow)
- Folio calibration: Latest month set to 26.12 Cr active folios

3. ETL Design & Architecture

The ETL pipeline follows a structured Extract-Transform-Load architecture. Raw CSV files are extracted, cleaned and validated, then loaded into a SQLite database and processed CSV files for downstream analytics.

3.1 Pipeline Architecture

Stage	Tool	Description	Output
Extract	Python/Pandas	Load raw CSVs from data/raw/	Raw DataFrames
Validate	Python	Schema checks, null detection	Validation Report
Transform	Pandas	Clean, normalize, scale, deduplicate	Cleaned DataFrames
Load	SQLite	Write to SQLite DB and processed CSVs	database.db + CSVs
Analyze	Python/Jupyter	Risk metrics, cohorts, recommender	Charts + CSVs
Visualize	Power BI	4-page interactive dashboard	mf_Dashboard.pbix

3.2 Data Cleaning Steps

Step 1: Fund Master

- Removed duplicate amfi_codes
- Parsed launch_date to datetime format
- Rounded expense_ratio_pct and exit_load_pct to 2 decimal places

Step 2: NAV History

- Removed null NAV values (coerce conversion errors)
- Deduplicated on (amfi_code, date) composite key
- Sorted by amfi_code and date for time-series analysis

Step 3: AUM Data

- Scaled latest month AUM to match ■81L Cr industry benchmark
- Added aum_lakh_crore derived column for readability

Step 4: Investor Transactions

- Removed 247 rows with null amount_inr values
- Stripped whitespace from city_tier and kyc_status columns
- Parsed transaction_date to datetime

3.3 SQLite Database Schema

1.1 SQLite Database Schema

Table	Primary Key	Foreign Keys	Rows
fund_master	amfi_code	—	40
nav_history	amfi_code + date	amfi_code → fund_master	46,000
scheme_performance	amfi_code	amfi_code → fund_master	40
investor_transactions	transaction_id	amfi_code → fund_master	32,778
portfolio_holdings	amfi_code + stock	amfi_code → fund_master	322
benchmark_indices	index_name + date	—	8,050
industry_aum	date	—	90
category_inflows	category + month	—	144

2. EDA Findings

Exploratory Data Analysis was performed across all 10 datasets to uncover patterns, trends, and anomalies in the Indian Mutual Fund industry.

2.1 Industry AUM Trend

The Indian Mutual Fund industry has shown consistent growth from ■39L Cr in March 2022 to ■81L Cr in December 2025 — a 107% growth in 3 years. The growth accelerated post-COVID as retail investors shifted from bank FDs to mutual funds.

Period	AUM (L Cr)	Growth YoY	Key Driver
Mar 2022	■38.96	—	Post-COVID recovery
Mar 2023	■41.77	+7.2%	Retail SIP growth
Mar 2024	■56.48	+35.2%	Bull market rally
Mar 2025	■70.32	+24.5%	Record SIP inflows
Dec 2025	■81.00	+15.2%	NFO launches + equity rally

2.2 SIP Inflow Trend

Monthly SIP inflows have grown from ■11,517 Cr in January 2022 to a record ■31,000 Cr in December 2025. This represents 169% growth and reflects the deepening of retail investor participation in equity markets.

- Average monthly SIP growth rate: 4.2% YoY
- Highest inflow month: December 2025 at ■31,000 Cr
- Correlation with Nifty 50: 0.78 (strong positive)

2.3 Transaction Analysis

Transaction Type	Count	Total Amount	Avg Amount	% Share
Lumpsum	~12,500	■2.3 Bn	■18,400	58.5%
Redemption	~11,500	■1.4 Bn	■12,174	35.3%
SIP	~8,778	■0.24 Bn	■10,997	6.2%

2.4 Geographic Distribution

Transaction amounts vary significantly across states, reflecting India's economic concentration:

- Punjab leads with highest transaction volume — ■0.32 Bn
- Tamil Nadu and Madhya Pradesh follow closely

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Tier-1 cities account for 68% of total transaction value

• Tier-3 cities show fastest growth rate at 23% YoY

2.6 Investor Demographics

Age Group	Avg SIP Amount	Preferred Category	Risk Profile
18-25	■ 8,200	Index Funds	Low-Moderate
26-35	■ 13,400	Small/Mid Cap	High
36-45	■ 15,800	Flexi Cap	Moderate-High
46-55	■ 12,100	Large Cap	Moderate
56+	■ 9,800	Debt/Liquid	Low

3. Performance Analysis

Comprehensive performance metrics were computed for all 40 mutual fund schemes, covering absolute returns, risk-adjusted returns, and benchmark comparisons.

3.1 Return Analysis

Category	Avg 1Y Return	Avg 3Y Return	Avg 5Y Return	Benchmark 3Y
Small Cap	24.5%	22.4%	21.1%	19.8%
Mid Cap	20.1%	18.9%	17.5%	17.2%
Flexi Cap	17.8%	15.4%	14.9%	15.0%
Large Cap	12.4%	12.0%	11.8%	11.5%
ELSS	18.2%	16.1%	15.3%	15.0%
Liquid	7.2%	6.5%	6.1%	6.0%
Gilt	8.1%	7.4%	7.0%	7.1%

3.2 Risk Metrics

Risk was measured using multiple indicators including Sharpe Ratio, Standard Deviation, Beta, and Value at Risk (VaR).

Fund Category	Sharpe Ratio	Std Dev	Beta	VaR (95%)	Risk Grade
Liquid	7.68	0.8%	0.02	-0.12%	Low
Gilt	2.41	4.2%	0.08	-0.68%	Low
Large Cap	1.06	14.2%	0.89	-1.62%	Moderate
Flexi Cap	0.94	16.8%	0.92	-1.98%	Moderate
Mid Cap	0.88	19.5%	1.02	-2.24%	High
Small Cap	0.82	24.1%	1.12	-2.71%	Very High
ELSS	0.91	17.2%	0.95	-2.05%	Moderate

3.3 Top Performing Funds by Sharpe Ratio

	Fund Name	Category	Sharpe	3Y Return	Risk Grade
1 3.4	ICICI Pru Liquid Fund - Regular	Liquid	7.68	6.5%	Low
2	ABSL Liquid Fund - Regular	Liquid	7.41	6.3%	Low
3 3.7	HDFC Top 100 Fund - Direct	Large Cap	1.06	13.8%	Moderate
4 3.8	Mirae Asset Large Cap - Regular	Large Cap	1.04	12.4%	Moderate
5	Axis Bluechip Fund - Direct	Large Cap	0.98	12.1%	Moderate

NAV vs Benchmark Analysis

NAV performance was indexed to 100 at the start date (Jan 2022) and compared against the Nifty 50 benchmark. Key observations:

i) Large Cap funds closely track Nifty 50 with minimal tracking error (0.3-0.5%)

ii) Small Cap funds outperform during bull markets by 8-12% annually

iii) Liquid funds maintain consistent upward trend with minimal volatility

iv) Index funds (ETFs) have lowest tracking error at 0.05-0.1%

2) Dashboard Overview

A 4-page interactive Power BI dashboard was built to visualize all key metrics. Each page focuses on a specific analytical dimension with full cross-filtering and drill-through capabilities.

a) Page 1 — Industry Overview

Visual	Type	Data Source	Purpose
Total AUM Card	KPI Card	07_scheme_performance	Show ■81L Cr AUM
SIP Inflows Card	KPI Card	04_monthly_sip_inflows	Show ■31K Cr SIP
Total Folios Card	KPI Card	06_industry_folio_count	Show 26.12 Cr folios
Total Schemes Card	KPI Card	07_scheme_performance	Show 1,908 schemes
AUM Trend	Line Chart	03_aum_by_fund_house	2022-2025 growth
AUM by AMC	Bar Chart	03_aum_by_fund_house	Top AMC comparison

b) Page 2 — Fund Performance

Visual	Type	Fields Used	Purpose
Risk-Return Scatter	Scatter Plot	return_3yr, std_dev, aum	Risk vs return
Fund Scorecard	Table	scheme_name, returns, sharpe	Sortable fund data
NAV vs Benchmark	Line Chart	nav_indexed, nifty_indexed	Performance compare
Fund House Slicer	Slicer	fund_house	Filter by AMC
Category Slicer	Slicer	category	Filter by type
Plan Slicer	Slicer	plan	Direct/Regular

c) Page 3 — Investor Analytics

Visual	Type	Fields Used	Purpose
Txn by State	Bar Chart	state, amount_inr	Geographic distribution
Txn Type Split	Donut Chart	transaction_type, amount_inr	SIP/Lump/Redeem split
Age vs SIP	Bar Chart	age_group, amount_inr avg	SIP by demographics
Monthly Volume	Line Chart	transaction_date, count	Transaction trends
State Slicer	Slicer	state	Geographic filter
City Tier Slicer	Slicer	city_tier	Tier 1/2/3 filter

d) Page 4 — SIP & Market Trends

Visual	Type	Fields Used	Purpose
SIP vs Nifty	Dual-Axis	sip_inflow_crore, nifty50_close	Correlation analysis
Category Heatmap	Matrix	category, month, net_inflow	Inflow heatmap
Top 5 FY25	Bar Chart	category, net_inflow_crore	Best FY25 categories

3) Advanced Analytics (Day 6)

a) Historical VaR & CVaR

Value at Risk (VaR) at 95% confidence was computed for all 40 funds using the historical simulation method. CVaR (Conditional VaR) measures the expected loss in the worst 5% of scenarios.

Risk Category	VaR (95%)	CVaR (95%)	Interpretation
Low Risk (Liquid/Gilt)	-0.12% to -0.68%	-0.18% to -0.92%	Very safe for capital preservation
Moderate (Large Cap)	-1.62% to -1.98%	-2.1% to -2.5%	Suitable for 3-5 year horizon
High (Mid/Small Cap)	-2.24% to -2.71%	-2.9% to -3.4%	Long term investors only (5+ yr)

b) Rolling 90-Day Sharpe Ratio

The rolling 90-day Sharpe Ratio was computed for the top 5 funds. Formula: rolling(90).mean() / rolling(90).std() x sqrt(252). This shows how risk-adjusted performance changes over time.

i) ICICI Pru Liquid Fund: Consistently high Sharpe (6-9 range)

ii) Large Cap funds: Sharpe peaks during bull markets (2023-2024)

iii) Small Cap funds: Volatile Sharpe — ranges from 0.2 to 2.1

c) Investor Cohort Analysis

Cohort Year	Investors	Avg SIP Amount	Total Invested	Top Fund
2024	4,624	■ 10,997	■ 21.5 Cr	Axis Bluechip Fund
2025	138	■ 13,505	■ 0.86 Cr	HDFC Top 100 Fund

d) SIP Continuity Analysis

Critical Finding: Only 2.2% of investors (30 out of 1,362) maintain regular SIP cadence. 1,332 investors are flagged as At-Risk with avg gap > 35 days.

e) Sector HHI Concentration

The Herfindahl-Hirschman Index measures portfolio concentration. HHI > 1500 = High concentration (sector risk). Axis Bluechip has the highest HHI at 2,064.

4) Limitations

While this project provides comprehensive analytics, there are certain limitations that should be considered when interpreting the findings:

#	Limitation	Impact	Mitigation
1	Sample Size: Only 40 of 1,908 scheme analyzed	Partial industry view	Scale with AMFI API
2	Synthetic Data: Some values scaled to match benchmarks	Minor accuracy impact	Use live AMFI data feed
3	No Live Data: Static CSVs used, no real-time update	Dashboard not live	Connect Power BI Service
4	SIP Continuity: Based on transaction gaps ,not mandates	Overestimates at-risk investors	Integrate mandate data
5	HHI: Based on single portfolio date Snap shorts	Not time-series	Monthly portfolio data
6	Geographic: State-level only, no district granularity	limited regional insights	Add PIN code mapping
7	Investor Age: Derived from age_group buckets ,not DOB	Approximate demographics	Use actual DOB field

a) Data Assumptions

i) Industry AUM scaled to ■81L Cr to match AMFI published figures for March 2025

ii) SIP inflow for December 2025 set to ■31,000 Cr based on AMFI projections

iii) Folio count for latest month set to 26.12 Cr based on AMFI reports

iv) Nifty 50 values for months beyond August 2025 are estimated projections

5) Recommendations

a) For Bluestock Fintech Platform

Priority	Recommendation	Expected Benefit
High	SIP Nudge System: Alert the 1,332 at-risk investors via SMS/email	15–20% SIP retention improvement
High	Live Data Integration: Connect to AMFI API fo real-time NAV updates	Accurate live dashboard
High	Fund Recommender: Deploy recommender.py as a web a	Personalized UX
Medium	Mobile Dashboard: Publish Power BI to mobile app	Broader reach
Medium	Expand to 1,908 schemes: Scale analytics to full fund universe	Complete market view
Low	Add ESG Scoring: Integrate ESG ratings for sustainable investing	Premium feature

b) For Investors

i) Low Risk Investors: Liquid and Short Duration funds offer best risk-adjusted returns (Sharpe 7.68)

ii) Moderate Risk: Large Cap funds provide steady 12-14% returns with manageable drawdowns

iii) High Risk: Small Cap funds for 5+ year horizons only — expect -2.7% daily VaR

iv) All Investors: Maintain SIP regularity — gaps > 35 days significantly reduce wealth creation

v) Young Investors (18-25): Start with ■500/month SIP in Index funds — low cost, market returns

c) For AMCs (Fund Houses)

i) Focus retention programs on the 2024 cohort — largest and most valuable investor base

ii) Reduce expense ratios for Direct plans to compete with growing passive fund adoption

iii) Launch more Flexi Cap and Multi-Asset funds to meet diversification demand

iv) Invest in Tier-3 city distribution — fastest growing market segment

6) Conclusion

This capstone project successfully demonstrates the power of data analytics in the mutual fund domain. Over 7 days, a complete analytics pipeline was designed and implemented — from raw data ingestion to professional-grade dashboards and advanced risk models.

The Indian Mutual Fund industry is at an inflection point — ■81L Cr AUM, ■31K Cr monthly SIP, and 26.12 Cr folios signal massive retail participation. Data analytics is the key differentiator for platforms like Bluestock.

a) All Objectives Met

Objective	Deliverable	Status
Data Ingestion & ETL	data_ingestion.py + SQLite DB	■ Complete
Data Cleaning	10 cleaned CSVs	■ Complete
EDA & Visualization	eda_visualization.py + charts	■ Complete
Power BI Dashboard	mf_Dashboard.pbix (4 pages)	■ Complete
Advanced Analytics	Advanced_Analytics.ipynb	■ Complete
Risk Metrics	var_cvar_report.csv + charts	■ Complete
Fund Recommender	recommender.py	■ Complete
Final Report	Final_Report.pdf (17 pages)	■ Complete
Presentation	Bluestock_MF_Presentation.pptx	■ Complete
GitHub Repository	v1.0 tag + README.md	■ Complete

b) Final Deliverables Summary

i) Final_Report.pdf — This document (17 pages)

ii) Bluestock_MF_Presentation.pptx — 12-slide presentation

iii) mf_Dashboard.pbix — 4-page Power BI dashboard

iv) Advanced_Analytics.ipynb — Jupyter notebook with all risk analytics

v) var_cvar_report.csv — VaR and CVaR for all 40 funds

vi) recommender.py — Interactive fund recommender system

vii) GitHub Repository — Complete code + README + v1.0 tag

Thank you for reviewing this project.

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